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CHEMICAL REACTION ENGINEERING (CHE 321)

MIDTERM EXAM 1

CLOSED BOOK, TIME 70 MIN

The homogeneous gas-phase decomposition of phosphine

$r_A = -k C_A$

$\frac{1\ mol}{h\, dm^3}$

X

$$4\, PH_3\,(g) \rightarrow P_4\,(g) + 6\, H_2\,(g)$$

$$4\, A \rightarrow B + 6\, C$$ → A is L.R.

$1 A \rightarrow \frac{1}{4} B + \frac{3}{2} C$

proceeds at 922.15 K with the first-order rate:

$T = 922.15\ K$

$X = 80\%$

$$\boxed{\begin{array}{c} r_A = -k\, C_A \\ k = 10\ hr^{-1} \end{array}}$$

$F_{A0} = 40\ mol/h$

15 1. What size of plug flow reactor (PFR), operating at constant temperature and pressure of 922.15 K and 460 kPa, can achieve $\boxed{80\%}$ conversion of a feed of 40 moles of pure phosphine per hour?

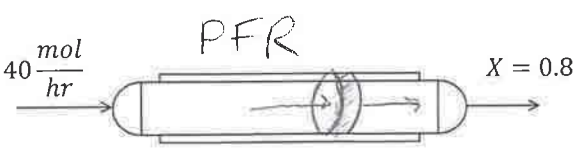
Conversion $X \equiv \frac{F_{A0}-F_A}{F_{A0}}$. Gas constant $R = 8.314\ \frac{Pa.m^3}{mol.K}$. $\int dx\, \frac{1+\varepsilon x}{1-x} = -(1+\varepsilon)\ln(1-x) - \varepsilon x$.

5 1.1. Based on a shell mole balance, derive a differential equation for conversion as a function of PFR volume. There is no pressure drop along the reactor

10 1.2. Express the rate of reaction in terms of conversion X.

1.3. Integrate the differential mole balance and determine the volume.

$V = \int_0^{X_1} -\frac{F_{A0}}{r_A(X)}\, dX$



$X = \frac{F_{A0} - F_A}{F_{A0}}$

1.1) Shell mol bal: $0 = F_{in} - F_{out} - V \cdot r_A$

$\frac{dV}{dX} = -\frac{F_{A0}}{r_a} \Rightarrow V = \int_0^{X_1} \frac{F_{A0}}{-r_a}\, dX$

1.2) $r_a = \left[\frac{mol}{m^3 \cdot h}\right] = -k C_A \rightarrow$ as a function of X $\Rightarrow = -k C_{A0} X_1$

$\frac{1\,mol\ C_A}{dm^3}$

$1\,mol\ A$

1.3) $V = F_{A0} \int_0^{X_1} \frac{1}{-r_a(x)}\, dx = 40\,\frac{mol}{h}\left(\frac{2}{r_a(x)^2}\right) = 12.5\ dm^3$

$10\,(0.8)^2$

$-r_a(x)^{-1}$

$+2\, r_a(x)^{-2} \rightarrow \frac{2}{r_a(x)^2}$

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